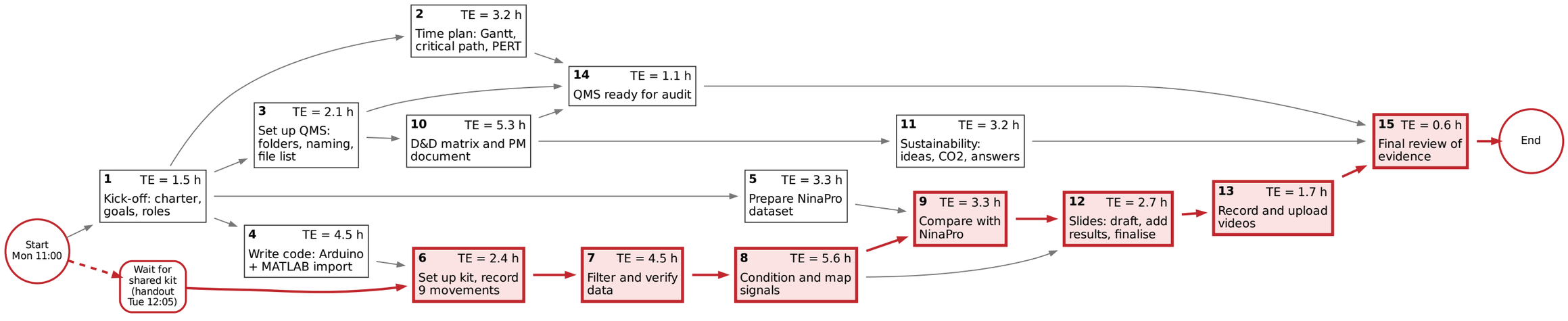
PW1 Group 32 ElectroMyo Gang - PERT chart V2

Activity-on-node network. Red = critical path. Times are working hours (08:00-20:00).



Three-point estimates (hours)

ID	Activity	Pred.	O	M	P	TE	σ	Slack	Critical
1	Kick-off: charter, goals, roles	-	1.5	1.5	1.5	1.50	0.00	7.1	
2	Time plan: Gantt, critical path, PERT	1	2	3	5	3.17	0.50	27.5	
3	Set up QMS: folders, naming, file list	1	1.5	2	3	2.08	0.25	21.1	
4	Write code: Arduino + MATLAB import	1	3	4	8	4.50	0.83	7.1	
5	Prepare NinaPro dataset	1	2	3	6	3.33	0.67	20.8	
6	Set up kit, record 9 movements	4, kit	1.5	2	5	2.42	0.58	0.0	Yes
7	Filter and verify data	6	3	4	8	4.50	0.83	0.0	Yes
8	Condition and map signals	7	3.5	5	10	5.58	1.08	0.0	Yes
9	Compare with NinaPro	8, 5	2	3	6	3.33	0.67	0.0	Yes
10	D&D matrix and PM document	3	4	5	8	5.33	0.67	21.1	
11	Sustainability: ideas, CO2, answers	10	2.5	3	5	3.25	0.42	21.1	
12	Slides: draft, add results, finalise	8, 9	2	2.5	4	2.67	0.33	0.0	Yes
13	Record and upload videos	12	1	1.5	3	1.67	0.33	0.0	Yes
14	QMS ready for audit	2, 3, 10	0.75	1	2	1.12	0.21	23.2	
15	Final review of evidence	11, 13, 14	0.5	0.5	1	0.58	0.08	0.0	Yes

How the estimates were used

TE = (O + 4M + P) / 6

σ = (P − O) / 6

σ total = √(Σ σ²) along the critical path

Critical path

Start → kit handout (Tue 12:05) → 6 → 7 → 8 → 9 → 12 → 13 → 15 → End

Expected finish

Kit wait 13.1 h + critical activities 20.8 h = 33.8 h → Thu 08:50

σ total = √(0.58² + 0.83² + 1.08² + 0.67² + 0.33² + 0.33² + 0.08²) = 1.7 h

68% chance of finishing between Wed 19:08 and Thu 10:31 (33.8 ± 1.7 h).